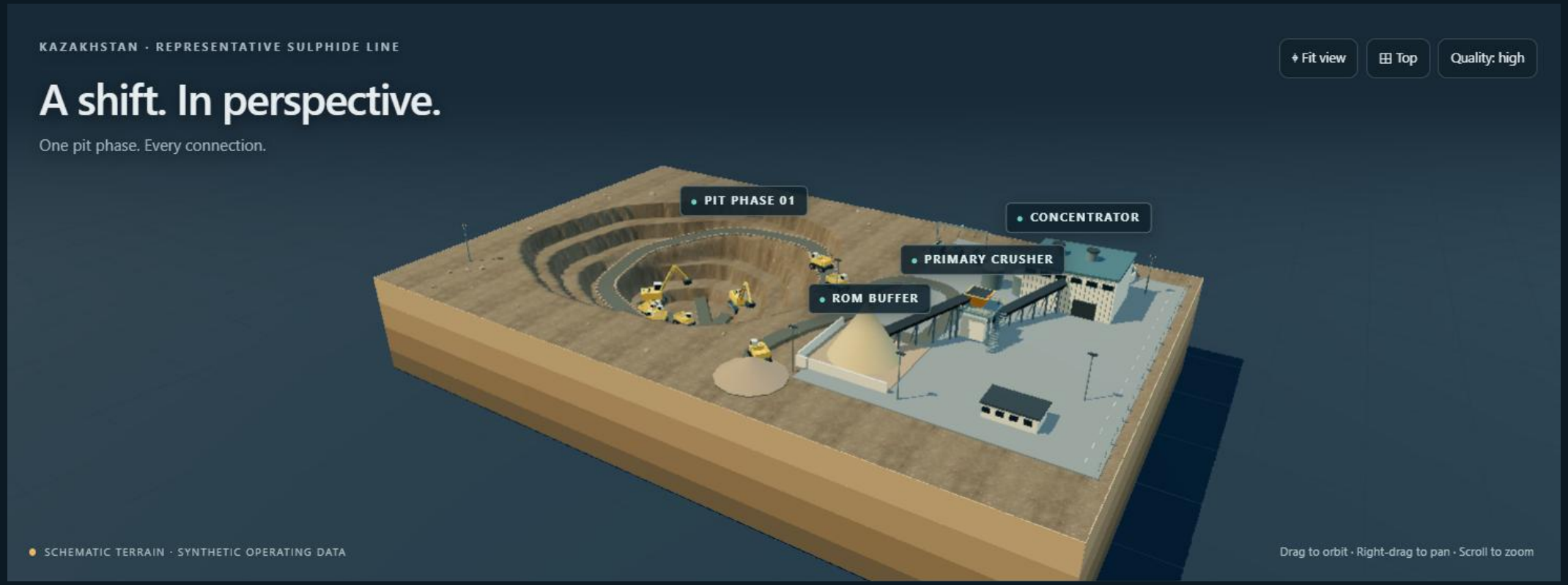
AKTOGAY

Sulphide Shift Twin



Tau-Ken Labs (Team-064)

Majhabul Hoque Tahsim • Sole member & developer

A real mine. A focused model.

Aktogay, Kazakhstan

Real-world reference

Open-pit copper operation with sulphide processing. The project represents one production line.

Model boundary

Pit faces, haulage, ROM stockpile, primary crusher and concentrator feed.

25 Mt/year

Representative line design basis

Synthetic operating data
Schematic terrain
No site calibration

The shift decision

How can the dispatcher keep plant feed steady after a disruption?

Dispatcher

Reassign trucks as equipment and haul routes change.

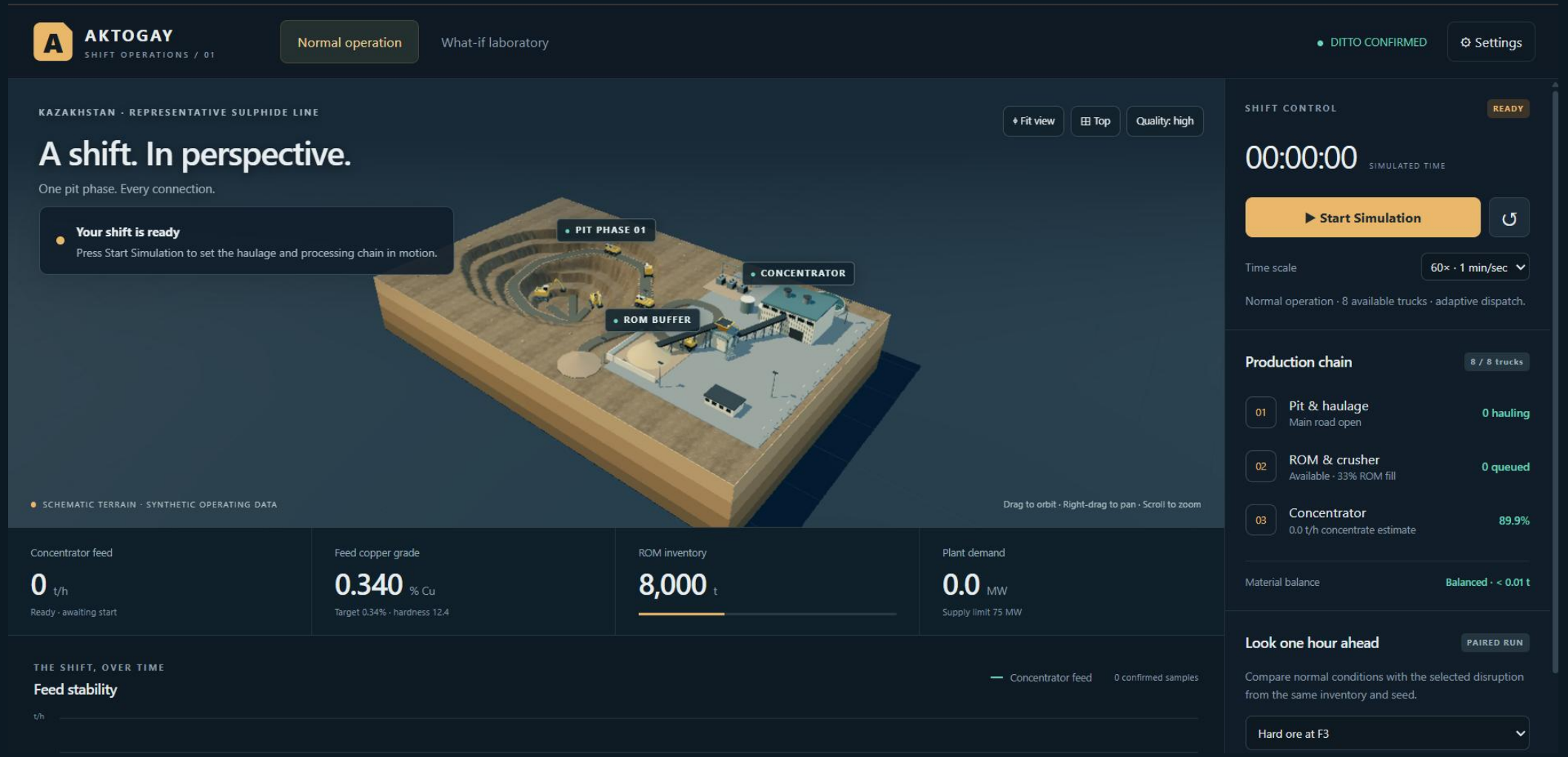
Mining engineer

Balance copper grade and ore hardness across available faces.

Plant supervisor

Track incoming tonnes, inventory and power constraints.

The operator workspace



One view connects equipment, shift controls and production KPIs.

Normal operation in motion



3,145

t/h concentrator feed

0.341%

Cu feed grade

66.0 MW

Synthetic snapshot

A controllable shift

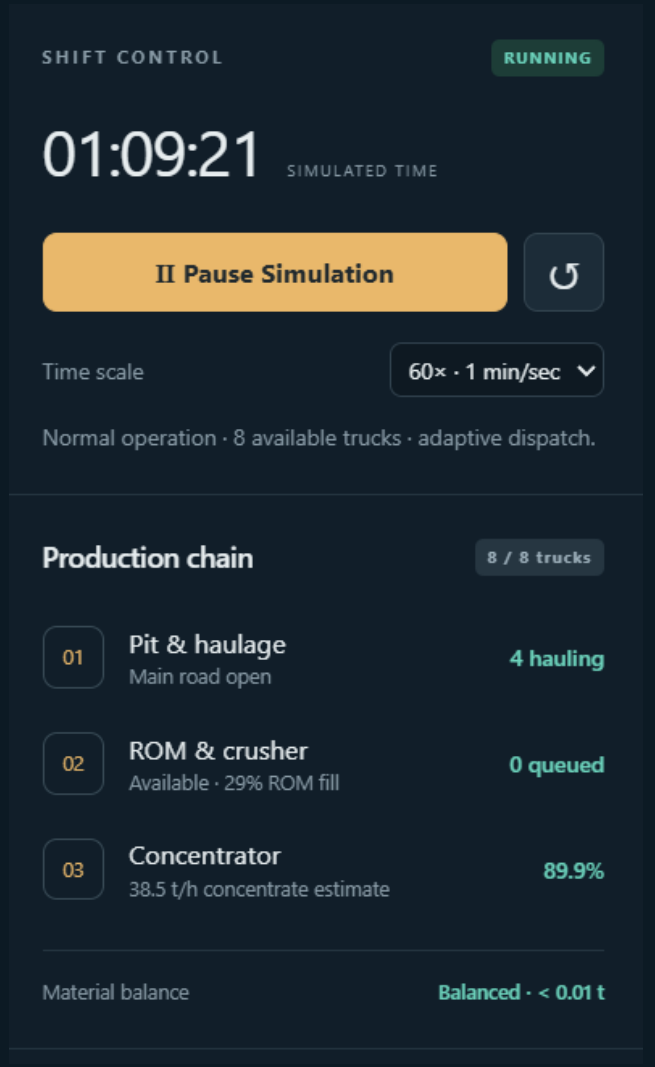
Start, pause and reset

Run the model deliberately, pause to inspect an event, or begin a fresh shift.

Adjust the pace

The captured run uses 60× speed: one simulated minute per second.

Visible material balance: < 0.01 t



Six what-if scenarios

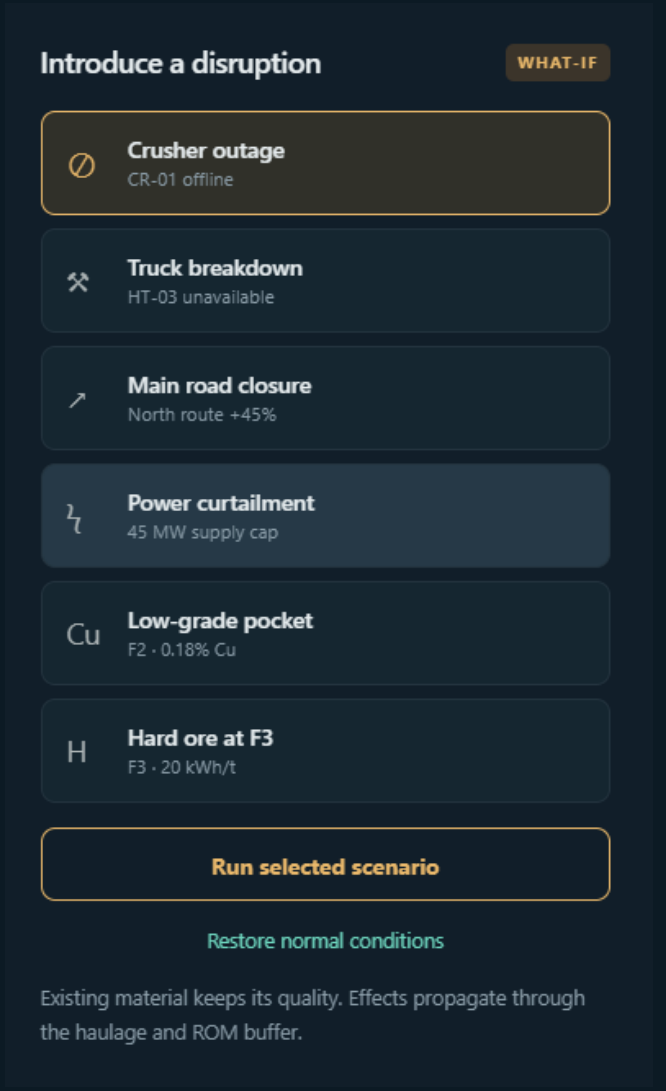
Equipment and routes

Crusher outage, truck breakdown and main road closure change flow and cycle time.

Power and ore quality

Power curtailment, low-grade F2 and hard F3 expose energy and blend constraints.

Select a scenario, run it, then restore normal conditions.



Crusher outage stops plant feed



0 t/h feed

8,317 t ROM inventory

Fault visible in the trend

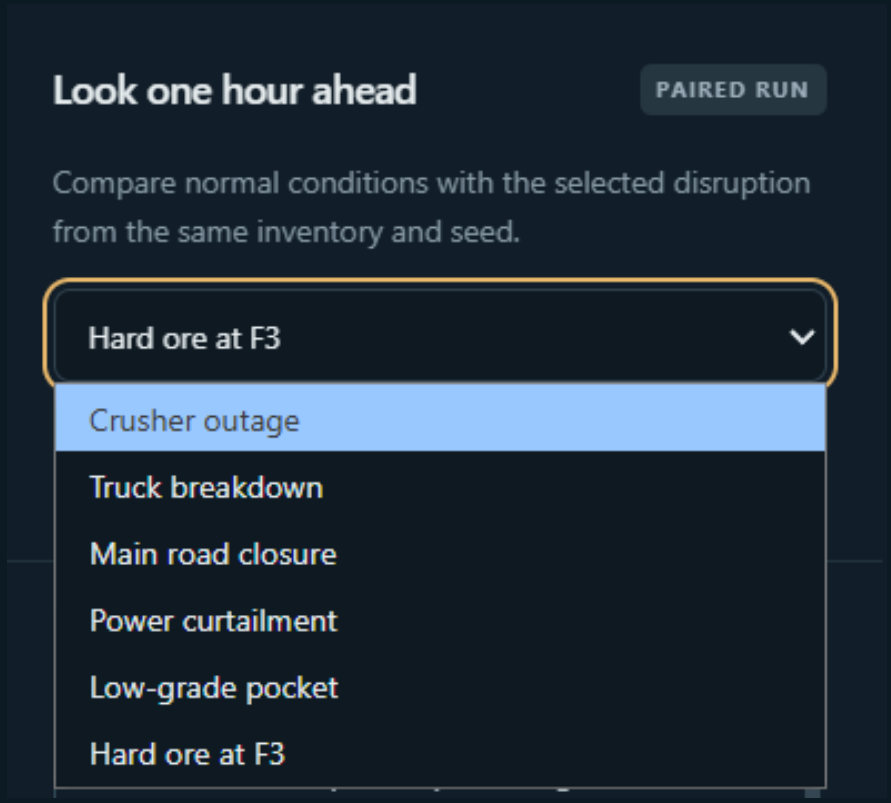
A one-hour decision experiment

Same starting state. Two possible shifts.

Baseline versus disruption

Compare normal conditions with the selected fault over the next 60 simulated minutes.

Inventory and random seed stay matched.



Settings change the actual model

Operating capacity

Fleet size, crusher throughput and power supply set the available resources.

Material and timing

Ore grade, hardness and cycle assumptions shape the downstream response.

Pause before applying a new shift setup.



3D exploration and graphics

The spatial view follows the simulation state



Inspection

Orbit, pan, zoom and select equipment to understand its current state.

Rendering choices

Fit or top view, selectable quality and locally bundled graphics assets.

One authoritative operator snapshot

01 Simulation model

Truck cycles, queues, ROM mixing, processing and energy constraints.

02 MQTT and Eclipse Ditto

Publish state and read the confirmed operator snapshot back.

03 3D operator interface

Clock, vehicles, KPIs and events share the same confirmed state.

04 Telemetry history

Telegraf, InfluxDB and Grafana support the platform’s data pipeline.

Freshness checks identify stale data instead of presenting it as live.

Evidence and engineering boundaries

21

model/API tests reported passed

17

local browser checks reported passed

Evidence in this deck

Actual developer screenshots show a running shift and a crusher-outage response.

Remaining boundary

Synthetic calibration. Full platform audit and judge-machine performance still need confirmation.

Tau-Ken Labs

A working shift twin for Aktogay’s production chain

Live demo: normal operation, disruption, recovery

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